INCH-POUND
A-A-60023A
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SUPERSEEDING
A-A-60023
June 8, 2022

COMMERCIAL ITEM DESCRIPTION

AUTOMATED MANUFACTURING MACHINE, COMPUTER NUMERIC CONTROL LATHE, FLATBED, HORIZONTAL SPINDLE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numeric control (CNC), horizontal-spindle, flatbed lathes and turning centers intended for turning, boring, drilling, reaming, tapping, and facing of workpieces of cast iron, steel, titanium, aluminum, brass, graphite epoxy, and composite materials for use in Aircraft Metals Technology Shops across the Air Force. The equipment identified herein shall be manufactured in the United States of America.

CNC lathe machining is an automated machine process used to manufacture engineering parts by selectively and progressively removing material from a workpiece. Material is removed (milled) using a rotating multi-point cutting tool and computerized controls (i.e., CNC). CNC-milled parts are most practically manufactured using Computer-Aided Design (CAD) and Computer-Aided Manufacturing (CAM) software. CAD enables the development and design of a part. CAM programs are used to generate manufacturing tool paths and automate the manufacturing processes. CAM converts the CAD model into an executable CNC program (NC) that the machine uses (via “G-code”) to manufacture the physical part.

G-code, per Electronic Industries Alliance (EIA) RS-274, is the required programming language; it describes geometric paths using a simple command structure of the letter “G” followed by a numeric value.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, Bldg. 28, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3416

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2. CLASSIFICATION

The flatbed lathes and turning centers shall be of the sizes specified in Table I. The size is the maximum cutting length (also referred to as the center-to-center distance) as shown in Table I. The size to be furnished shall be as specified (see 7.2.b).

TABLE I. Sizes of Flatbed CNC Lathes and Turning Centers.

Size	Chuck Size	Max Speed	Max Spindle Rating
30.0 in	8.0 in	1800 rpm	10 hp
48.0 in	8.0 in	1800 rpm	10 hp

3. SALIENT CHARACTERISTICS

3.1 General requirements. The lathe or turning center shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The lathe or turning center shall be constructed from materials with sufficient mass and strength and configured in a shape and size that shall support the workpiece and machine components while minimizing element deflections and initiated and/or transmitted vibration during operation. The rotational axis of the workpiece shall be horizontal. The longitudinal axes of machine ways supporting the carriage, tailstock, steady rests, or tool rests shall lie in a horizontal plane and shall be parallel to the rotational axis of the workpiece. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Access doors or removable panels shall be provided for servicing all components located within the lathe or turning center.

The machining center and CNC system shall be shipped to the AF ready to operate pending training and installation. The term "ready to operate" is defined herein as a minimum set of tooling and accessories, internal and external hardware and software, and components required for installation to operate the machining center and CNC systems to perform milling operations.

3.2 Components. The lathe or turning center shall include, at a minimum, the following components:

3.2.1 Base. The base shall support the lathe or turning center bed and all components mounted thereon. The base shall provide a means for leveling and anchoring the machine. A swarf-collection area shall be provided to collect and retain turning chips and coolant fluid either adjacent to or within the base.

3.2.2 Base Anchoring. The lathe or turning center shall be delivered with fasteners, adapters, clamps, or other hardware required to affix the base to the floor. Maximum diameter for each hole required in the shop floor shall be 1 inch.

3.2.3 Bed. The bed shall support the machine ways upon which the carriage, tailstock, and tool or workpiece rests travel. The bed shall be configured to allow the chip swarf to either pass through openings in the bed or be diverted to the collection area.

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3.2.4 Spindle and Spindle Drive. The spindle shall be fabricated from alloy steel, then hardened and ground. The spindle shall be supported on preloaded, antifriction roller or ball bearings within the headstock. All spindle bearings shall be sealed and lubricated for life. The spindle shall be dynamically balanced both before and after assembly. The spindle shall be thermally isolated from the headstock and spindle drive. The spindle nose shall conform to American Society of Mechanical Engineers (ASME) B5.9. Spindles shall be compatible with provided chucks (see section 3.2.5) as listed in Table I Spindles shall conform to the requirements listed in Table II.

TABLE II. Spindle Requirements.

Size	Max Speed	Bar Capacity	Chuck Size	Drive Type	Peak Power[1]
30.0 in	1800 rpm	2.30 in	8.0 in	Belt	108 ft-lb @ 335 rpm
48.0 in	1800 rpm	2.30 in	8.0 in	Belt	108 ft-lb @ 335 rpm

Note 1: Peak power equates to 200 percent spindle load for three (3) minutes, 150 percent spindle load (75 percent of shown value) for 15 minutes, or 100 percent spindle load (50 percent of shown value) continuously.

3.2.5 Chuck. The lathe or turning center shall have a chuck provided in the size noted in Table I. The chuck shall be a 3-jaw chuck featuring serrated jaws and clamping force necessary to hold the workpiece during turning operations. Adapters and hardware necessary to mount the chuck must be included. Lathes or turning centers shall include a manual, wrench-operated chuck. Chucks shall conform to American Society of Mechanical Engineers (ASME) B5.9.

3.2.6 Machine ways. Machine ways shall be the standard design of the manufacturer. A means shall be provided to prevent coolant and chip swarf from accumulating on machine ways and becoming lodged between sliding parts. Machine ways shall be machined/ground smooth to allow continuous, uninterrupted movement of carriages, guides, and slides for machining operations. Machine ways shall not corrode or degrade under normal operations and maintenance conditions and/or storage for no less than fifteen (15) years from the date of procurement.

3.2.7 Carriage. Carriage(s) shall be mounted on the bed machine ways and shall move parallel to the rotational axis of the workpiece. Carriages shall support movement and use of the cross-slide(s). Carriages shall be a configuration allowing the movement of the cross-slide or tool turret assembly to be on a plane parallel to the spindle axis. Carriages shall be capable of programmable, hands-free movement.

3.2.8 Cross-slide and Tool Turret Assembly. The lathe or turning center shall have a cross-slide mounted on the carriage. The cross-slide shall allow for tool travel perpendicular to the rotating axis of the workpiece. The cross-slide shall support the functionality of the applicable tool station and tool turret. The cross-slide and tool turret shall be capable of programmable, hands-free movement.

3.2.9 Tool turret(s). A tool turret shall be provided with a minimum of four (4) stations for automated tool changing and indexing for unattended operations. The lathe or turning center

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shall be operable with the tool turret installed or removed. Tool turrets shall have the capability of programmable, hands-free changing between installed tools.

3.2.10 Axis drive mechanisms. Each motion-control axis of the lathe or turning center shall be actuated by an independent axis drive. Each axis drive shall have a brushless servomotor coupled directly onto a ball screw anchored at both ends and a closed-loop feedback control system to position the driven component.

3.2.11 Tailstock. The lathe or turning center shall have a removable tailstock located on the bed opposite the spindle. The centerline of the tailstock shall be coincident with the centerline of the spindle. The tailstock shall be equipped with a moveable quill with a minimum 6 inches of travel and shall accommodate various material-holding centers and cutting tools. The tailstock shall have a locking device to hold it in position after adjustment.

3.2.12 Safety devices. An emergency stop switch/button shall be provided that when pressed will instantly stop all motion of the machine, servomotors, tool changers, and coolant pump. The system shall allow the operator to define the maximum allowable tool load, as a percentage, and the resulting action when that load is exceeded, including system alarm. Limit switches shall be provided at both ends of each axis of travel to prevent over-travel. Automatic machine-actuated safety warnings, trips, and/or stoppages shall activate an indicator light on the operator control panel and produce a visual alarm. Torque-limiting capabilities shall be included on all axis drives. The lathe or turning center shall have a protective enclosure with locking doors that prevent access to the workpiece and tools when in locked operation. When in unlocked operation, the rapid motion shall be limited to 25 percent of normal rapid operations, spindle rotation shall be limited to 50 rpm, and tool change shall not be allowed. The enclosure shall be configured to afford easy operator access to the workpiece when doors are open and clear observation when the lathe or turning center is in use.

3.2.13 Control system. The lathe or turning center shall be capable of manual control from a local operator station, semi-automatic control from manual data input (MDI) devices, and automatic control from programs stored in the memory of the CNC system or external drives. The control-system components shall be industrial grade, and enclosures shall be type 12 in accordance with National Electrical Manufacturers Association (NEMA) 250. The control system shall have the capability to perform a screen capture to save an image of the current control screen to the hard drive or external USB drive. The lathe or turning center shall be capable of operating in the following modes:

a. Manual operation mode.

b. Manual data input (MDI) mode.

c. Programming mode.

d. Automatic mode.

e. Background editing mode.

f. Machine diagnostic/maintenance/monitoring modes.

g. Remote monitoring using an Android or iOS device.

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h. Wireless file transfer using an Android or iOS device.

3.2.13.1 Machine-status indicators. The lathe or turning center shall include, as a minimum, the following machine-status indicators:

a. Power-on indicator.

b. Operational mode indicator.

c. Spindle-power indicator.

d. Spindle-speed indicator.

e. Axis-movement direction indicators.

f. Axis-movement speed indicators.

g. Accessory system (e.g., air, hydraulic, coolant, lubrication) status indicator(s).

h. Overload and/or trouble indicator(s).

i. Overload and/or trouble alarm.

3.2.13.2 Manual controls. The lathe or turning center manual controls shall include, as a minimum, the following functions:

a. Emergency stop.

b. Key-actuated system lock.

c. Power on/off.

d. Spindle start/stop.

e. Spindle-speed override.

f. Carriage motion.

g. Cross-slide motion.

h. Tailstock motion.

i. Tool-station indexing.

j. Feed hold.

k. Feed-rate override.

l. Jog motion and jog direction.

m. CNC-system start/stop (cycle start/stop).

3.2.13.3 Computer numeric control. The lathe or turning center shall have a CNC system that is a completely integrated, functional unit. The CNC system shall operate as a continuous-path system providing full contouring capability. The term “full contouring”, as used herein, shall be defined as a closed-loop control system having simultaneous control of position, direction, and rate of movement of all controlled axes. The CNC system shall always be active when electric power is supplied to the lathe or turning center. The CNC system shall, as a minimum, automatically control the following functions:

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a. Spindle speed.

b. Carriage motion.

c. Cross-slide motion.

d. Tool-station indexing.

e. All accessory systems (e.g., air, hydraulic, coolant, lubrication).

f. Other machine functions considered by the manufacturer as standard.

3.2.13.4 Hardware features. The CNC system shall include, as a minimum, the following hardware features:

a. Minimum of three (3) 32-bit microprocessor having a minimum operating speed of 1000 blocks per second.

b. Buffer-storage capacity sufficient to allow continuous program execution without non-programmed stops, delays, or pauses.

c. A USB port capable of reading external USB devices formatted in FAT32.

d. A minimum 15-inch, full-color, LCD touchscreen allowing screen navigation, data entry, and full operational control.

e. Minimum 1 GB internal memory for program/operational storage.

f. Standard Ethernet port for network connectivity.

g. Wi-Fi connectivity.

3.2.13.5 Software features. The CNC system shall include, as a minimum, the following programming software and control features:

a. EIA RS-274 variable data-block programming format (standard G-codes).

b. Decimal-point programming.

c. Conversational mode programming.

d. Inch/metric programming.

e. Programmable zero-point reference.

f. Absolute and incremental motion-control programming.

g. Simultaneous control of all motion axes.

h. Linear, circular, and helical motion interpolation.

i. Constant surface-speed capability.

j. Manufacturer's standard suite of preprogrammed machine operational cycles.

k. Workpiece graphic display.

l. Tool-path graphic display.

m. Tool-length compensation.

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n. Tool offsets.

o. Scale-feedback compensation.

p. Drive-backlash compensation.

q. Maintenance-monitoring package.

r. Diagnostics assistance.

s. Support user-defined macros (i.e., subroutines for repeated cycles, probing routines, special functions, etc.).

3.2.13.6 Data Package. The CNC system shall include an external data package with maintenance and technical support that shall provide, as a minimum, the following hardware and software:

a. Three-year multi-user subscription and/or license to the most current version of MasterCAM (see 7.4c).

b. Three-year subscription and/or license to the most current version of SOLIDWORKS Professional design and modeling software.

c. Personal computer (PC) with Windows 10 or higher operating system, multiple core 64-bit processor, 16GB RAM, 2GB Open FL 2.0-complaint graphics card, 1920 x 1200 screen resolution, and 16GB disk space.

d. All hardware and software shall be capable and compatible with machine controllers and post-processors.

3.2.14 Compressed-air system. A compressed-air system connected either to shop air or to a dedicated air compressor and tank shall be provided. Connection ports to and from the compressed air system shall be 3/8-inch supplied at 80 psi. The compressed-air system shall include piping, valves, fittings, regulated water/oil separator, and other appurtenances necessary for the operation of the pneumatic components. The compressed-air system shall conform to the requirements of NFPA T2.25.1M.

3.2.15 Electrical system. The electrical system shall conform to the National Fire Protection Association (NFPA) 79. The system shall have multiple electrical connectivity options per Table III.

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TABLE III. Power Requirements.

Size	Power Option 1	Power Option 2	Power Option 3
30.0 in	220 VAC 25 A Three Phase	440 VAC 13 A Three Phase	220 VAC 40 A Single Phase
48.0 in	220 VAC 25 A Three Phase	440 VAC 13 A Three Phase	220 VAC 40 A Single Phase

3.2.16 Electric motors. The spindle-drive motor and motion-control motors shall be brushless, sealed, and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to NEMA MG 1 and NEMA ICS 16, respectively. The spindle-drive motor shall have an open drip-proof enclosure. Motion- and position-control motors and controls shall conform to NEMA ICS 1 and NEMA ICS 3.

3.2.17 Tooling, Attachments, and Accessories. All tooling, attachments, accessories, and hardware required to render the machine usable shall be included, including, as a minimum:

3.2.17.1 Toolholders. A minimum of two removable toolholders shall be provided. Toolholders shall be functional with the lathe or toolholder provided and able to be fully integrated with the CNC and software. The toolholder shall be capable of holding C-style inserts via top clamping and hole clamping, though not simultaneously. The toolholder shall be in a right-handed, external tool orientation and have a 95-degree angle. The toolholder shall conform to ISO 13399 standards.

3.2.17.2 Standard Boring Bar. A removable boring bar shall be provided that is functional with the lathe or toolholder and can be fully integrated with the CNC and software. The boring bar shall be capable of holding C-style inserts via top clamping and via hole clamping. The toolholder shall be in a right-handed, external tool orientation and have a 95-degree angle. The toolholder shall conform to ISO 13399 standards. The boring bar shall have a length of 8 inches and a minimum machining diameter of 1.2 inches.

3.2.17.3 Fine Finish Inserts. A minimum of ten C-style inserts shall be provided with the appropriate grade for fine finishing of steel and cast iron. Inserts shall be compatible with the toolholder specified in 3.2.17.1. Insert sizes shall be 0.5 inches with a maximum thickness of 0.19 inches.

3.2.17.4 Medium Finish Inserts. A minimum of ten C-style inserts shall be provided with the appropriate grade for medium-light finishing of steel and cast iron. Inserts shall be compatible with the toolholder specified in 3.2.17.1. Insert sizes shall be 0.5 inches with a maximum thickness of 0.19 inches.

3.2.17.5 Steady rest. A removable steady rest shall be provided for increased support when turning long or narrow workpieces. The steady rest shall be able to be positioned anywhere along

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the workbed in the Z direction and mechanically locked into place. The steady rest shall be usable with or without the tailstock in use. The steady rest shall be adjustable to support workpieces from a minimum of 0.25 inches up to 5.0 inches.

3.2.17.6 Quick-Change Tool Kit. A removable quick-change tool post shall be provided that allows for quick and repeatable manual tool changes. The kit shall include all hardware for mounting the tool post onto the system, at least four turn/face toolholders compatible with the toolholders specified in 3.2.17.1, and a boring-bar holder compatible with the boring bar specified in 3.2.17.2. The tool post shall be capable of rotating/adjusting orientation and locking into place.

3.3 Equipment Training. A complete overview of the machine controller, graphics display, and machine function and operation shall be provided at the installation location. Training shall be coordinated with the training point of contact (POC) at the installation location no later than seven calendar days after setup/commissioning at the installation location. The training POC will be provided by the Government at the time of order. Training shall follow installation of the equipment and last no more than three days for a maximum of five technicians. Training shall provide the necessary skills and knowledge to operate the machine safely, execute manual data-input functions, manually operate machinery, load tooling into tool changer, conduct tooling offsets, and importing manufacturing files. Electronic versions of training materials shall be accessible to training participants for a minimum of three years after completion of the training.

3.4 Warranty. A minimum one-year warranty shall be provided to protect against defects, breaks, or component failures during normal operations. The warranty shall include the replacement and installation of components/assemblies at the installation location by an OEM-certified technician. The warranty does not cover consumable items such as tooling identified in 3.2.17.2 through 3.2.17.4. For equipment located in austere operating environments, an OEM-certified technician may oversee the repair by a Government representative; however, replacement components shall still be provided as part of the warranty.

3.5 Support. Technical support for the equipment and software shall be provided for a period of three years after the date of purchase. Technical support shall be accessible, at a minimum, via phone and email.

3.6 Performance. When tested in accordance with section 5.3, the lathe or turning center shall meet either the precision or standard performance category requirements shown in Table IV.

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TABLE IV. Performance requirements.

Test	Parameter	Performance category	
		Precision	Standard
Linear displacement accuracy	All X, Y, and Z directions	0.0003 inch/10 inches	0.001 inch/10 inches
Bidirectional repeatability	All X, Y, and Z directions	±0.0002 inch	±0.0005 inch
Maximum parallelism error between spindle C-axis and each Z-axis		0.0003 inch	0.0005 inch
Spindle dynamic error	Radial error-average	0.0003 inch	0.0005 inch
Spindle thermal growth		0.0015 inch	0.0025 inch
Spindle C-axis indexing accuracy		±1.8 arc seconds	±3.6 arc seconds
Spindle C-axis indexing repeatability		±1.0 arc second	±1.8 arc seconds
Turret indexing accuracy		±3 arc seconds	±5 arc seconds
Turret indexing repeatability		±1.5 arc seconds	±2.5 arc seconds
Contouring performance-ball bar test	100 degrees, 190 degrees and 360 degrees	0.0005 inch	0.0015 inch

3.7 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.6. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall comply with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.6 and OSHA standards, the OSHA standards shall apply.

3.8 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. Measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.9 Nameplate. A nameplate shall be securely attached to the machine and shall contain the following information:

a. Nomenclature.

b. Manufacturer's name.

c. Serial number.

d. Machine model designation.

e. Power input (volts, total amperes, phase, and frequency).

f. Short-circuit/overcurrent rating.

g. Contract number or order number.

h. Applicable National stock number.

i. Date of manufacture.

3.10 Lubrication plate or chart. A lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.

b. Servicing interval.

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c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE

5.1 Product conformance. The products shall meet the salient characteristics (section 3) of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The lathe or turning center shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. The lathe or turning center shall be tested in accordance with sections 5.3.1 through 5.3.1 Should a malfunction or a failure occur during any test, the problem shall be corrected, and the test shall be rerun from the beginning.

5.3.1 Test setup and conditions. The manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to the National Institute of Standards and Technology (NIST) and shall provide certification of calibration. The machine shall be tested in the environmental conditions considered by the manufacturer as standard for machine-performance verification.

5.3.2 Operational test. The lathe or turning center shall be operated in accordance with the manufacturer's standard operating procedures. The lathe or turning center shall be operated for not less than four hours in which proper operation of all controls and features shall be verified.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with procedures considered by the manufacturer as standard for machine performance verification.

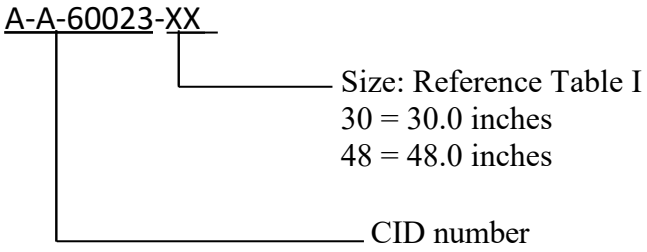
5.4 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the performance requirements shown in table IV. Failure of the machine to meet the performance category requirements shown in table IV shall be cause for rejection.

6. PACKAGING Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

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7.1 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



7.2 Sources of referenced documents.

7.2.1 The American Society of Mechanical Engineers (ASME). Copies of ASME documents may be obtained at https://www.asme.org/.

7.2.2 American National Standards Institute (ANSI). Copies of ANSI documents may be obtained at https://www.ansi.org.

7.2.3 Electronic Industries Association (EIA). SAE International (formerly named the Society of Automotive Engineers). Copies of SAE documents may be obtained at: https://www.sae.org/.

7.2.4 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/.

7.2.5 National Electrical Manufacturers Association (NEMA). Copies of NEMA standards may be obtained at https://www.nema.org/.

7.2.6 National Fire Protection Association (NFPA). NFPA documents may be obtained at https://www.nfpa.org/.

7.2.7 International Organization for Standardization (ISO). Copies of ISO Standards may be obtained at https://www.iso.org/home.html.

7.2.8 Occupational Safety and Health Administration (OSHA). Copies of OSHA documents may be obtained at https://www.osha.gov.

7.3 Applicable National Stock Numbers (NSNs). The following NSNs are established to meet the requirements of the associated PINs:

PIN	NSN
A-A-60023-30	3416-01-698-8834
A-A-60023-48	3416-01-698-9410

7.4 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.

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b. Size of lathe or turning center required (see 2).

c. Three (3) year multi-user subscription and/or license to most current version of MasterCAM software.

d. Point of contact for training coordination at installation location (name, address, email, and telephone numbers).

7.5 Key Words.

Boring
Drilling
Reaming
Turning

MILITARY INTERESTS:

Custodian:
Air Force – 184

Preparing Activity:
Air Force – 184

Reviewer:
GSA FAS

Agent:
Air Force – 110

(Project 3416-2024-001)

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